

Scaling Book Exercises: Section 12, Quiz 1

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Quiz: GPU Hardware

Source scan: ../handwritten/section_12_handwritten.pdf, pages 1–3

Reference: <https://jax-ml.github.io/scaling-book/gpus/>

Question 1: CUDA cores

An H100 has 132 streaming multiprocessors (SMs), four subpartitions per SM, and 32 fp32 CUDA cores per subpartition:

$$132 \cdot 4 \cdot 32 = 16,896 \text{ CUDA cores.}$$

A B200 has 148 SMs:

$$148 \cdot 4 \cdot 32 = 18,944 \text{ CUDA cores.}$$

A TPU v5p has two TensorCores, four independent ALUs per VPU lane, and $8 \cdot 128$ lanes per TensorCore:

$$2 \cdot 4 \cdot 8 \cdot 128 = 8,192 \text{ ALUs.}$$

The TPU therefore has roughly half as many independent vector ALUs as an H100.

Question 2: H100 vector FLOPs

Ignoring fused multiply-add counting, the base-clock vector throughput is

$$132 \cdot 4 \cdot 32 \cdot 1.59 \times 10^9 = 26.9 \text{ TFLOPs/s.}$$

At the 1.98 GHz boost clock it is

$$132 \cdot 4 \cdot 32 \cdot 1.98 \times 10^9 = 33.5 \text{ TFLOPs/s.}$$

The H100 has 990 bf16 Tensor-Core TFLOPs/s, about $30\times$ the non-FMA vector rate.

Question 3: Matmul intensity

The critical arithmetic intensity is peak matmul throughput divided by HBM bandwidth:

$$I_{\text{H100,fp16}} = \frac{990 \times 10^{12}}{3.35 \times 10^{12}} = 295 \text{ FLOPs/byte},$$
$$I_{\text{B200,fp16}} = \frac{2250 \times 10^{12}}{8 \times 10^{12}} = 281 \text{ FLOPs/byte}.$$

FP8 doubles the matmul throughput, giving approximately

$$590 \text{ FLOPs/byte on H100,} \quad 562 \text{ FLOPs/byte on B200.}$$

Question 4: B200 matmul runtimes

For $\text{fp16}[64, 4096] \times \text{fp16}[4096, 8192]$, the effective batch $64 < 281$, so the operation is bandwidth-bound. It moves

$$2BD + 2DF + 2BF = 2(64)(4096) + 2(4096)(8192) + 2(64)(8192) \approx 69 \times 10^6 \text{ bytes.}$$

The ideal runtime is therefore

$$T \approx \frac{69 \times 10^6}{8 \times 10^{12}} = 8.6 \mu\text{s},$$

or roughly 10–12 μs in practice.

For batch 512 > 281 , the matmul is compute-bound:

$$T \approx \frac{2(512)(4096)(8192)}{2.25 \times 10^{15}} \approx 15.3 \mu\text{s},$$

with roughly 20 μs being a practical expectation.

Question 5: H100 fast-memory capacity

Each of the 132 SMs has 256 kB of SMEM and 256 kB of register memory:

$$132(256 \text{ kB}) \approx \boxed{33 \text{ MB of SMEM}}$$

and independently about 33 MB of registers, or 66 MB combined. A modern TPU has about 120 MB of VMEM but only 256 kB of vector registers in total.

Question 6: B200 clock frequency

The 18,944 CUDA cores can perform two FLOPs per cycle with an FMA:

$$18,944 \cdot 2 = 37,888 \text{ FLOPs/cycle.}$$

Thus the clock implied by 80 vector fp32 TFLOPs/s is

$$f = \frac{80 \times 10^{12}}{37,888} \approx 2.1 \text{ GHz.}$$

Question 7: H100 vector-add runtime

Adding two fp32 vectors of length N performs N FLOPs. Two inputs are read and one output is written, moving

$$4N + 4N + 4N = 12N \text{ bytes.}$$

Hence

$$I = \frac{N}{12N} = \frac{1}{12} \text{ FLOP/byte.}$$

The two roofline times are

$$T_{\text{math}} = \frac{N}{33.5 \times 10^{12}}, \quad T_{\text{comms}} = \frac{12N}{3.35 \times 10^{12}} = \frac{N}{2.79 \times 10^{11}}.$$

Because $T_{\text{comms}} \gg T_{\text{math}}$, vector addition is strongly memory-bound. For $N = 1024$, the ideal full-device roofline gives

$$T_{\text{math}} \approx 0.031 \text{ ns}, \quad T_{\text{comms}} \approx 3.67 \text{ ns}.$$

Only 1024 CUDA cores, or about eight SMs, can be occupied at this size, so the compute estimate would be roughly $16\times$ larger even before kernel launch latency. In practice, fixed latency dominates both estimates.

For $N = 1024^3$,

$$T_{\text{math}} \approx 32.0 \text{ } \mu\text{s}, \quad \boxed{T_{\text{roofline}} \approx T_{\text{comms}} \approx 3.85 \text{ ms.}}$$

No empirical GPU measurement is claimed in the handwritten work.